1 Ancestry of a sample of genes

1.1 Motivating Example: human evolution and the infinitely-many-sites model

One of the signature early applications of the coalescent was to inference regarding the early history of humans. Several of the earliest data-sets consisted of short regions of mitochondrial DNA [mtDNA] or Y chromosome. Since mtDNA is maternally inherited it is perfectly described by the original version of the coalescent, with its reliance upon the existence of a single parent for each individual and its recombination-free nature. To motivate what follows, here we consider one of those early data sets.

The data in the following example comes from Ward *et. al.* (1991). The data analysis and mathematical modeling comes from a paper by Griffiths and Tavaré (1994).

Mitochondria DNA (mtDNA) comprises only about 0.00006% of the total human genome, but the contribution of mtDNA to our understanding of human evolution far outweighs its minuscule contribution to our genome. Human mitochondrial DNA, first sequenced by Anderson *et.al.* (1981), is a circular double-stranded molecule about 16,500 base pairs in length, containing genes that code for 13 proteins, 22 tRNA genes and 2 rRNA genes. Mitochondria live outside the nucleus of cells. One part of the molecule, the control region (sometimes referred to as the D-loop), has received particular attention. The region is about 1,100 base pairs in length.

As the mitochondrial molecule evolves, mutations result in the substitution of one of the bases A,C,G or T in the DNA sequence by another one. Transversions, those changes between purines (A,G) and pyrimidines (C,T), are less frequent than transitions, the changes that occur between purines or between pyrimidines.

It is known that base substitutions accumulate extremely rapidly in mitochondrial DNA, occurring at about 10 times the rate of substitutions in nuclear genes. The control region has an even higher rate, perhaps on order of magnitude higher again. This high mutation rate makes the control region a useful molecule with which to study DNA variation over relatively short time spans, because sequence differences will be found among closely related individuals. In addition, mammalian mitochondria are almost exclusively maternally inherited, which makes these molecules ideal for studying the maternal lineages in which they arise. This simple mode of inheritance means that recombination is essentially absent, making inferences about molecular history somewhat simpler than in the case of nuclear genes.

In this example, we focus on mitochondrial data sampled from a single North American Indian tribe, the Nuu-Chah-Nulth from Vancouver Island. Based on the archaeological records (cf. Dewhirst, 1978), it is clear that there is a remarkable cultural continuity from earliest levels of occupation to the latest. This implies not only that there was no significant immigration into the area by other groups, but that subsistence pattern and presumably the demographic size of the population has also remained roughly constant for at least 8,000 years. Based on the current size of the population that was sampled, there are approximately 600 women of child bearing age in the traditional Nuu-Chah-Nulth population. The data consist of 55 individuals and 352 sites in the mitochondrial DNA. There are 18 segregating sites in all; 13 of the sites

	1	1	2	2	3			1	1	1	1	1	2	2	2	2	3	3	
Position	0	9	5	9	4	8	9	2	4	6	6	9	3	6	7	7	1	3	
	6	0	1	6	4	8	1	4	9	2	6	4	3	7	1	5	9	9	
Site	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Lineage Frequencies
Lineage																			
a	A	G	G	A	A	T	C	C	T	C	T	T	C	T	C	T	T	C	2
b	A	G	G	A	A	T	C	C	T	T	T	T	C	T	C	T	T	C	2
c	G	A	G	G	A	C	C	C	T	C	T	T	C	C	C	T	T	T	1
d	G	G	A	G	A	C	C	C	C	C	T	T	C	C	C	T	T	C	3
e	G	G	G	A	A	T	C	C	T	C	T	T	C	T	C	T	T	C	19
f	G	G	G	A	G	T	C	C	T	C	T	T	C	T	C	T	T	C	1
g	G	G	G	G	A	C	C	C	T	C	C	C	C	C	C	T	T	T	1
h	G	G	G	G	A	C	C	C	T	C	C	C	T	C	C	T	T	T	1
i	G	G	G	G	A	C	C	C	T	C	T	T	C	C	C	C	C	T	4
j	G	G	G	G	A	C	C	C	T	C	T	T	C	C	C	C	T	T	8
k	G	G	G	G	A	C	C	C	T	C	T	T	C	C	C	T	T	C	6
l	G	G	G	G	A	C	C	C	T	C	T	T	C	C	C	T	T	T	4
m	G	G	G	G	A	C	C	T	T	C	T	T	C	C	C	T	T	C	3
n	G	G	G	G	A	C	T	C	T	C	T	T	C	C	T	T	T	C	1

Figure 1: North American Indian tribe from Vancouver Island.

are pyrimidines and 5 are purines

The original data, appearing in Ward *et. al.* (1991) comprised a sample of mt DNA sequences from 63 individuals. The sample approximated a random sample of individuals in the tribe, to the extent to which this can be experimentally arranged. Each sequence is the first 360 basepair segment of the control region. The region comprises 201 pyrimidine sites and 159 purine sites; 21 of the pyrimidine sites are variable (or segregating), that is, not identical in all 63 sequences in the sample. In contrast, only if 5 of the purine sites are variable. There are 28 distinct DNA sequences (hereafter called lineages) in the data. Because, no transversions are seen in these data each DNA site is binary, having just two possible bases at each site.

To keep the presentation simple, we focus on one part of the data that seems to have a relatively simple mutation structure. We shall assume that substitutions at any nucleotide position can occur only once in the ancestry of the molecule. This is called **the infinitely-many-sites assumption.** Hence we have eliminated lineages in which substitutions are observed to have occurred more than once. The resulting subsample comprises 55 of the original 63 sequences, and 352 of the original 360 sites. Eight of the pyrimidine segregating sites were removed resulting in a set of 18 segregating sites in all; 13 of these sites are pyrimidines, and 5 are purines. These data are given in Table 1, subdivided into sites containing purines and pyrimidines. Each row of the table represents a distinct DNA sequence, and the frequency of these lineages are given in the right most column of the table.

What structure do these sites have? Because of the infinitely-many-sites assumption, the pattern of segregating sites tells us something about the mutations that have occurred in the

history of the sample. Next we consider an ancestral process that could have given rise the observed pattern of variability. This is called the coalescent.

The coalescent process, which we will discuss in some detail as the course goes on, is a way to describe the ancestry of the sample. The coalescent has a very simple structure. Ancestral lines going backward in time coalesce when they have a common ancestor. Coalescence occur only between pairs of individuals. This process may also be thought of as generating a binary tree, with the leaves representing the sample sequences and the vertices where ancestral lines coalesce. The root of the tree is the most recent common ancestor (MRCA) of the sample.

Example 1 To get an idea for building trees from sequences we begin with a simple example. Consider just the segregating purines of lineage b, c, d, e from Table 1. Below is this reduced data set.

	Site	1	2	3	4
lineage					
b		A	G	G	A
c		G	A	G	G
d		G	G	A	G
e		G	G	G	A

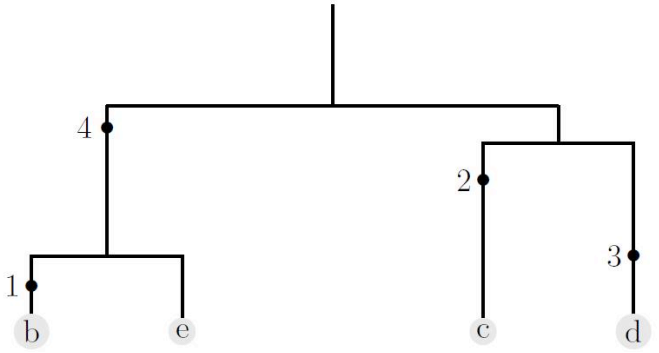


Figure 2: A tree consistent with the reduced data set for lineages b, c, d, e

Suppose $G\ G\ G\ G$ is the ancestral sequence to the four lineages. Figure 1 represents one possible evolutionary scenario connecting the individuals in the sample. The $\bullet$ represents a mutation. The number next to the $\bullet$ represents the position of the mutation. We read the tree diagram in Figure 1 as follows. Start at the ancestral tip of the tree. The first event to occur is a split in the ancestral line. Next a mutation occurs and a G mutates to an A at position 4. This mutation is passed on to lineage b and e. The tree splits again and a mutation occurs at site 2 in lineage c. Next, a mutation occurs at site 3 in lineage d. The final split in the tree separates lineage b and e. The last evolutionary event is a mutation at site 1 in lineage b.

A coalescent tree consistent with the Nuu-Chah-Nulth data set is given below

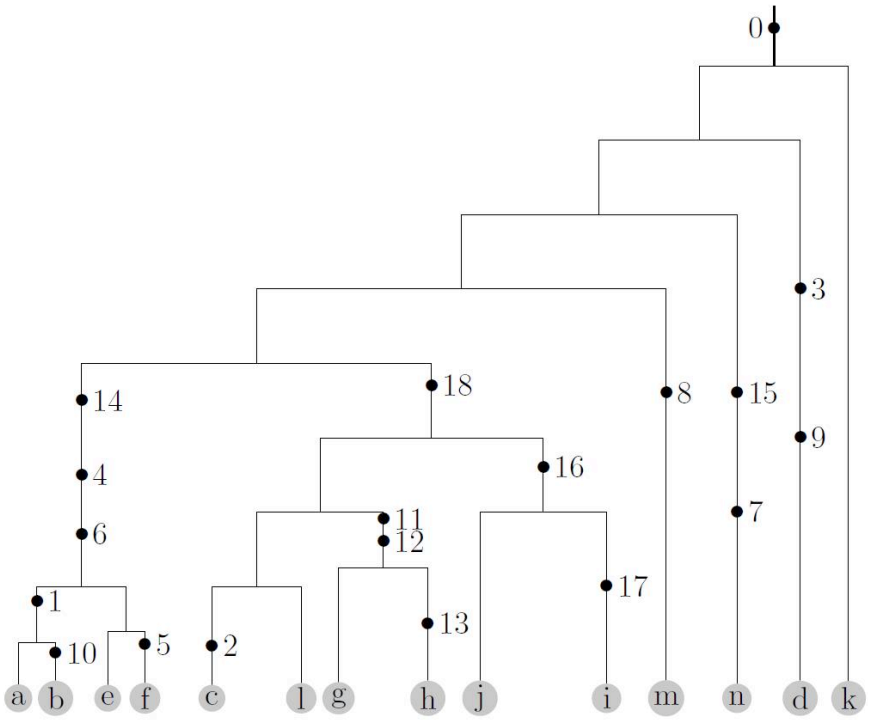


Figure 3: A tree showing the evolution of $n = 14$ lineages

Exercise 1: Convince yourself that Figure 3 represents a coalescent tree that is consistent with the data given in the table published by Ward et al 1991. Use the most frequently occurring basepair at each site as the ancestral sequence. Describe each event that lead to the sample. Construct another coalescent tree that is consistent with the data.

A rooted and unrooted gene tree consistent with the large table of 55 individuals shown above is given below

Exercise 2: Since mutations can occur only once in a given site, there is an ancestral type and a mutant type at each segregating site. For the moment assume we know which is which, and label the ancestral type is 0 and the mutant type as 1. To fix ideas, take each column of the data in Table 1 and label the most commonly occurring base as 0, the other as 1. Construct a matrix of 0's and 1's for the data in Table 1 in the manner described above.

The matrix of 0's and 1's can be represented by a rooted tree by labeling each distinct row by a sequence of mutations up to the common ancestor. These mutations are the vertices in the tree. This rooted tree is a condensed description of the coalescent tree with its mutations, and it has no time scale in it. Figure 4 represents a rooted condensed tree consistent with the data in table 1.

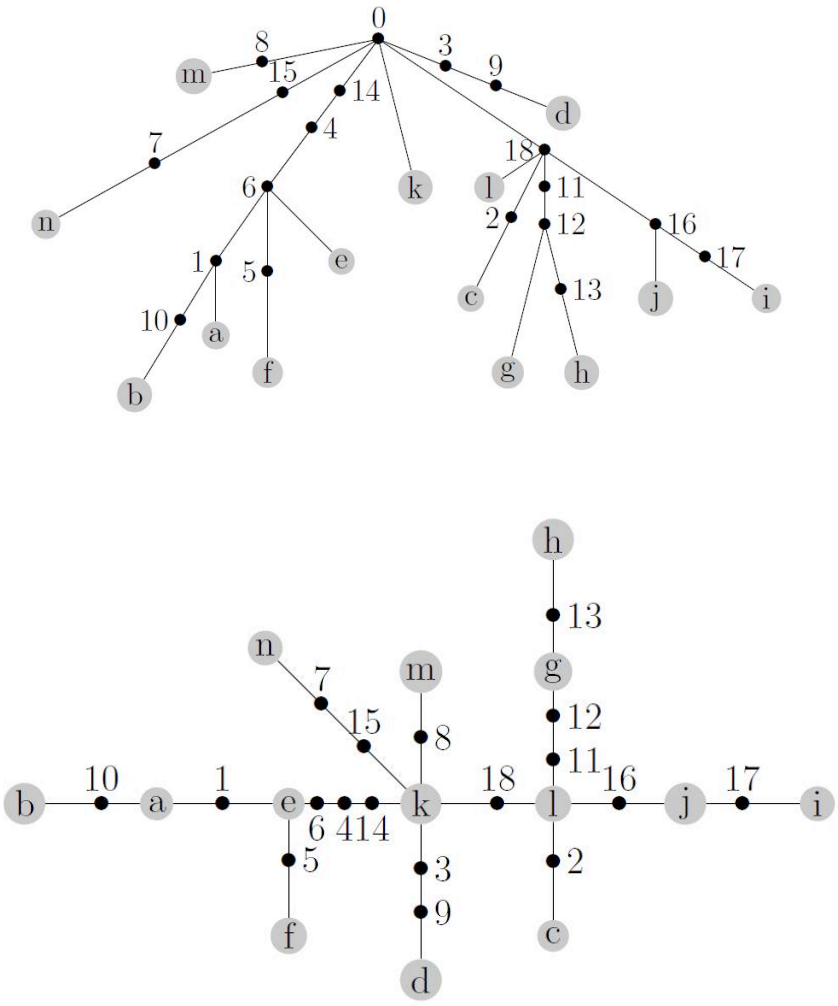


Figure 4: A rooted and an unrooted tree plotted using the data from the Nuhh-Chah-Nhulth tribe

Exercise 3 Verify that Figure 4 is consistent with the data. Take lineages a, b, c, d, j and draw the rooted condensed tree for this subset of individuals.

Of course, in practice we never know which type at a site is ancestral. All that can be deduced then from the data are the number of segregating sites between each pair of sequences. In this case the data is equivalent to an *unrooted* tree whose vertices represent distinct lineages and whose edges are labeled by mutations between lineages. The unrooted tree corresponding to the rooted tree in Figure 3a is shown in Figure 3b. All possible rooted trees may be found from an unrooted tree by placing the root at a vertex or between mutations, then reading off mutation paths between lineages and the root.

Exercise 4 Construct three rooted trees consistent with the unrooted tree in Figure 4.

1.2 Selectively neutral mutations in the coalescent process

The coalescent process wouldn't be an interesting process to study if it wouldn't allow including explicitly one of the most important biological processes for the genesis and maintenance of diversity: the mutational process. As we saw in the example of the mitochondrial DNA, there are various types of mutations. However, in the coalescent process we assume that all the mutation types follow an important assumption: neutrality, or absence of selection. In the absence of selection, the mutational process and the transmission of genes from one generation to the next are independent processes. That is, having information about one of these two processes doesn't tell us anything about the other. Furthermore, any gene, regardless of its type, has the same probability of leaving a descendant from one generation to the next. The three most important mutational models are the following: the **infinite alleles model**, the **infinite sites model** and the **finite sites model**. The first one assumes that the only information we have is whether any two alleles in the sample are identical or not. This model was developed as a response to the data gathered by electrophoresis analyses. It assumes that the analyst cannot have quantitative information about the differences between the haplotypes, that a mutation always create a new type and that it is impossible to know if what separates one individual form the other is one or more mutations. The second model, the infinite sites model, assumes that the molecular evolution occurs in very long chains with very low mutation rates at each nucleotide position. The mutations, according to this model, always "fall" in new positions (no "double hits" -*i.e.* at most one mutation per nucleotide position-) and that the amount of variable sites in the DNA is small with respect to the total number of sites. The third model allows for multiple hits and assumes a finite DNA sequence. There are many submodels arising from this model but we won't be covering them here.

1.3 Estimating the parameter θ

Let the total length of a coalescent tree is the sum of all the branch lengths and L_n. That is,

$$L_n = \sum_{j=2}^{n} jT_j,$$

where the random variables $T_j \sim \text{iid}\exp\left(\binom{j}{2}\right)$. Since the length of its branches are exponentially distributed random variables, it is possible to find analytically the distribution of L_n. However, for now let's just look at the mean and variance of the total length. To do that, let S_n denote the total number of mutations (segregating sites) in the genealogical tree up until the most recent common ancestor in a sample of genes of size n. Then, $S_n|L_n \sim \text{Poisson}\left(\frac{\theta L_n}{2}\right)$ and

$$
\begin{aligned}
E(S_n) &= E(E(S_n|L_n)) \\
&= E(\theta L_n/2) \\
&= \frac{\theta}{2} E\left(\sum_{i=2}^{n} iT_i\right) \\
&= \frac{\theta}{2} \sum_{i=2}^{n} iE(T_i) \\
&= \frac{\theta}{2} \sum_{i=2}^{n} i \frac{2}{i(i-1)} \\
&= \theta \sum_{j=1}^{n-1} \frac{1}{j}.
\end{aligned} \tag{1}
$$

Note that if n is large, then $E(S_n) \sim \log n$. In what follows, we display values of $E(S_n)$ for various sample sizes, when $\theta = 4$.

Sample size n	θ	$E(S_n)$
10	4	9.21
20	4	11.98
40	4	14.76
45	4	15.23
50	4	15.65
60	4	16.38
100	4	18.42

The variance of S_n, $\mathrm{Var}(S_n)$ informs us about how variable is the process of mutation accumulation for any realization depicting the evolution of a sample of n genes. Using the total variance formula, we see that the variance of S_n results from adding the variance due to the mutational process to the variance due to the genealogical process:

$$
\mathrm{Var}(S_n) = E(\mathrm{Var}(S_n|L_n)) + \mathrm{Var}(E(S_n|L_n)). \tag{2}
$$

The first term, $E(\mathrm{Var}(S_n|L_n))$, tells us on average, how does the variability of the mutational process contributes to the total variability in the tree. The second term, $\mathrm{Var}(E(S_n|L_n))$, quantifies how the mean number of mutations in the tree varies when the variation due to the coalescent process itself is taken into account. This term is directly proportional to the variance of the total tree length, which is completely determined by the coalescent process. Developing

each one of these two terms we get

$$\begin{aligned}
\text{Var}(S_n) &= E(\text{Var}(S_n|L_n)) + \text{Var}(E(S_n|L_n)) \\
&= E\left(\frac{\theta}{2}L_n\right) + \text{Var}\left(\frac{\theta}{2}L_n\right) \\
&= \frac{\theta}{2}E(L_n) + \frac{\theta^2}{4}\text{Var}(L_n) \\
&= \theta\sum_{i=1}^{n-1}\frac{1}{i} + \frac{\theta^2}{4}\text{Var}\left(\sum_{i=2}^{n} iT_i\right) \\
&= \theta\sum_{i=1}^{n-1}\frac{1}{i} + \frac{\theta^2}{4}\sum_{i=2}^{n} i^2\text{Var}\left(T_i\right) \\
&= \theta\sum_{i=1}^{n-1}\frac{1}{i} + \frac{\theta^2}{4}\sum_{i=2}^{n} i^2\frac{4}{i^2(i-1)^2} \\
&= \theta\sum_{i=1}^{n-1}\frac{1}{i} + \theta^2\sum_{i=1}^{n-1}\frac{1}{i^2}
\end{aligned}$$

When n is large,

$$\text{Var}(S_n) = \theta\log n + 2\theta^2.$$

Below is a table with the mean and standard deviation (squared-root of the variance) of S_n:

n	θ	$E(S_n)$	Stdev(S_n)
10	4	9.21	6.00
20	4	11.98	6.10
40	4	14.76	6.20
45	4	15.23	6.21
50	4	15.65	6.23
60	4	16.38	6.25
100	4	18.42	6.32

Now, what can we say about a population of interest, for a particular sample? That is, how can we estimate θ using the information in the sample about segregating sites? Let $a_n = \sum_{i=1}^{n-1}\frac{1}{i}$ and $b_n = \sum_{i=1}^{n-1}\frac{1}{i^2}$. Then, the mean and the variance of S_n can be written as:

$$E(S_n) = a_n\theta \quad \text{and} \quad \text{Var}(S_n) = a_n\theta + b_n\theta^2.$$

Define

$$\hat{\theta}_S = \frac{S_n}{a_n}.$$

This is the so called "Watterson estimator" of segregating sites (Waterson 1975). Note that, from the mean of S_n, we get that $E(\hat{\theta}_S) = \theta$. Statistical estimators that follow the property of having an average identical to the parameter they are estimating are called "unbiased estimators". Watterson's estimator is unbiased, whereas the Maximum Likelihood (ML) estimator arising from the infinite alleles model is not. Furthermore, from the variance of S_n we find that

$$\mathrm{Var}(\hat{\theta}_S) = \frac{1}{a_n}\theta + \frac{b_n}{a_n^2}\theta^2, \tag{3}$$

from which it fllows that $\mathrm{Var}(\hat{\theta}_S) \to 0$ as $n \to \infty$. The statistical estimators that follow this second property are known as "consistent" estimates. That means that one could attain any level of precision if the sample size is large enough. For this case, however, the size of the precision is not going to be much better than the size of the estimator, unless ridiculously large samples are taken.

There is another estimator that is very simple to understand, and that is also unbiased. It's "Tajima's D_n". If we consider that the definition of θ, is "the mean number of mutations that separate any two individuals taken at random from the population", then one simple thing we can do is to count the number of mutations that separate every pair of individuals in the sample and then average these numbers.

Let S_{ij} be the number of mutations that separate individuals i and j in the sample. Then,

$$D_n = \frac{2}{n(n-1)} \sum_{i \leq j} S_{i,j},$$

and since $E(S_{ij}) = E[S_2] = \theta$, it follows that

$$E(D_n) = \frac{2}{n(n-1)} \sum_{i \leq j} E(S_{ij}) = \frac{2}{n(n-1)} \sum_{i \leq j} E(S_{ij}) = \frac{2}{n(n-1)} \frac{n(n-1)}{2}\theta = \theta,$$

Tajima's estimator is therefore denoted as $\hat{\theta}_T = D_n$. It is possible to show that

$$Var(D_n) = \frac{n+1}{3(n-1)\theta} + 2\frac{(n^2+n+3)}{9n(n-1)}\theta^2.$$

If $n \to \infty$, the variance of this estimator converges to $\frac{\theta}{3} + \frac{2}{9}\theta^2 \neq 0$.

Question: Compute Tajima's D and Watterson's estimator of θ for the Nuu-Chah-Nulth population data set.